International Journal of Heat and Mass Transfer 107 (2017) 934–944

Contents lists available at ScienceDirect

International Journal of Heat and Mass Transfer

journal homepage: www.elsevier.com/locate/ijhmt

Numerical investigation on heat transfer and melting process of ice with different porosities

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ARTICLE INFO

Article history:
Received 12 July 2016
Received in revised form 19 October 2016
Accepted 1 November 2016
Available online 10 November 2016

Keywords:
Heat transfer
Natural convection
Ice melting
Porous medium
Enthalpy-porosity

ABSTRACT

The density of ice changes with icing conditions. The ice with different densities can be treated as a porous medium with different porosities. Porosity affects the ice melting process. A melting model based on the enthalpy-porosity method is applied to study the effects of porosity on the ice melting process and heat transfer. An electrothermal de-icing system is used to investigate the melting process of ice with different porosities. The transient process of ice melting is investigated, including the change of ice-water interface, the melted ice volume and the melted ice area on the surface. The effects of porosity on the temperature distribution of the ice-airfoil surface are analyzed. The effect of natural convection on ice melting process is also considered in this study. The results show that the porosity affects the temperature distribution of the ice melting zone, melting rate, melting ice area on the surface and melting ice volume significantly. The melting rate, melting ice area on the surface and melting ice volume in the ice melting process are all rise with the increase of porosity under the same heating condition. The investigation shows that the enthalpy-porosity method can describe the ice melting process more accurately.

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1. Introduction

Icing is a serious problem that influences the efficiency and safety of some equipment such as wind turbines and aircrafts. It usually occurs in cold regions when supercooled droplets in clouds are intercepted by wind turbine blades, aircraft wings, engine inlet, or other aircraft components. For wind turbine, icing affects the wind assessment phase and the operation of wind farms [1], reduces the efficiency and even causes damage [2,3]. For aircraft, icing affects the aerodynamic performance such as increasing drag and reducing lift. Under some extreme climate conditions, icing even causes crash [4]. According to the survey, many accidents happened due to icing problems. So, icing is a safety problem need to be solved. Nowadays, anti-icing or de-icing systems are widely used to alleviate or avoid icing phenomenon [5]. anti-icing systems can avoid the formation and accretion of ice, while de-icing systems are designed to remove the accreted ice. As a kind of de-icing system, the electrothermal de-icing system uses electric heat source to melt the ice accreted on the outer surface of icing components. The ice melts after the electrothermal de-icing system working for a period of time. The melting efficiency depends on two factors: one is the electric heat power and the other is the thermal physical property of ice. Investigate the effects of thermal physical properties on the ice melting process is helpful to design more suitable de-icing system. There are three different types of ice will formed on the icing components. They are defined as glaze ice, rime ice and mixed ice, respectively. The main difference of these three kinds of ice is their densities because of different porosity filled with air. Porosity affects the ice melting process under the same electric heating condition. The ice melting process involves many issues, such as melting zone temperature distribution, melting rate, melting ice area, melting ice volume.

At present, some researches have been devoted to the area of the heat transfer and melting process of ice. For heat transfer and melting process of ice, such as the temperature distribution and the motion of the interface was analyzed by means of a finite-difference method [6]. The melting process in a rectangular enclosure, driven by the coupling of heat conduction in the solid phase and natural convection in the phase-change material (PCM), was studied [7]. Three-dimensional melting of ice around a liquid-carrying tube placed in an adiabatic rectangular cavity is investigated by numerical analysis [8]. Heat transfer and melting process of ice in a spherical enclosure [9], in a rectangular enclosure [10,11] and in a rectangular enclosures with horizontal partial fins [12] were studied. The characteristics of ice melting heat transfer in a horizontal elliptical tube immersed in water [13] and heat transfer characteristics of lauric acid in a rectangular thermal storage unit

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http://dx.doi.org/10.1016/j.ijheatmasstransfer.2016.11.004
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Nomenclature

Symbol	Quantity (SI unit)
C	specific heat (J/(kg·K))
D	diameter (cm)
H	total enthalpy (J)
ΔH	latent heat (J/kg)
h	sensible enthalpy (J)
k	thermal conductivity (W/(m·K))
L	latent heat of a material (J/kg)
LWC	liquid water content (g/m^3)
MVD	mean volume diameter of water droplets (μm)
m	mass (kg)
PCM	phase-change material
Q	heat source (J)
S	environment function
T	temperature (°C)
V	volume (m^3)
v	velocity (m/s)
ρ	density (kg/m^3)
α	thermal diffusivity (m^2/s)
β	liquid fraction
$\varnothing$	porosity

Subscripts, superscripts	
ambient	ambient environment
eff	effective
f	fluid
fusion	fusion
ice	ice
inner	inner face
j	grid point in y-direction
l	liquid
liquidus	liquidus
lm	liquid at the melting point temperature
max	max
melt	melt
p	constant pressure
pore	pore
ref	reference
s	solid
sm	solid at the melting point temperature
solidus	solidus
total	total
wall	wall
x	space coordinate in the x direction
y	space coordinate in the y direction
z	space coordinate in the z direction

[14] were studied. The freezing and melting process of a small water droplet was investigated [15,16]. The turbulent convection beneath a horizontal ice-water interface was investigated by direct numerical simulation and laboratory experiments [17]. An experimental determination was made of the velocity profiles which result from the free convective melting of a vertical ice sheet into fresh water with temperature range from 2.0 °C to 7.0 °C [18]. The fluid flow under solid-liquid phase change conditions was studied numerically [19], experimental and theoretical study has been performed on the melting of spherical ice particles [20–23]. The effect of buoyancy-driven flow of cold water on the melting of a flat ice surface inside a porous matrix [24] and effect of thermal and mass diffusion in a cavity with differentially heated side walls [25] were studied. The melting driven convection at the ice-seawater interface was investigated by direct numerical simulation [26]. A natural convection model for melting phase change in porous media was established [27]. The melting process with natural convection in an inclined cavity has been studied using the enthalpy-based lattice Boltzmann method [28]. Previous research mentioned above, the ice melting process was studied when ice was treated as a constant density material rather than porous medium.

Although some research have been focused on the area of the heat transfer, melting process and density change of ice, little research focus on the effect of porosity on heat transfer and melting process of ice. The investigation is useful for improving the efficiency of de-icing system. In order to study the effect of porosity on heat transfer and melting process of ice, a numerical investigation is implemented in this paper.

This paper proposed a numerical method to analyze the melting process of ice with different porosity. The pore filled with air in the ice. The volume ratio of the pore and the ice is porosity. The buoyancy-driven flow in the melted zone as well as the natural convection heat transfer in the melted zone is also considered. The melting model with considering the ice density is used to simulate the ice melting process. The streamlines of natural convection in the melted zone were analyzed.

2. Theoretical model

2.1. Enthalpy method

When ice melts to water, two phases exist. The ice and the water are separated by the moving interface. The difficulty in simulating a phase change process is the presence of a moving boundary or region on which heat and mass balance conditions have to be met. The general method to solve the moving boundary problem is enthalpy method. The enthalpy method treats the enthalpy as a dependent variable and hence can write a single energy equation for the whole domain [29,30]. In the process of solving the energy equation, determination the interface position is not necessary in the enthalpy method. In ice melting model, the enthalpy is assumed to be a function of temperature. The enthalpy-temperature relationship curve is shown in Fig. 1. Where T_{melt} is the melting point of ice, H_{sm} and H_{lm} represent the enthalpy associated with the ice and water at the melting point, respectively.

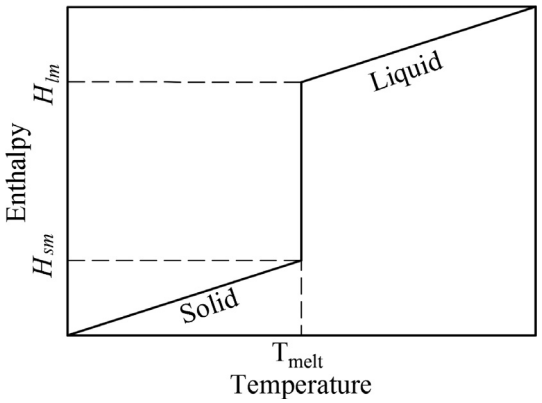


Fig. 1. Enthalpy-Temperature relationship graph.

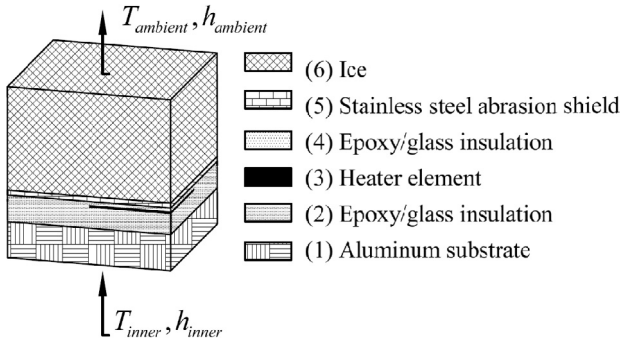


Fig. 2. Physical model of electrothermal de-icing system.

The enthalpy method can be used to simulate the ice melting process. A real physical model of ice melting is usually very complex and contains a lot of uncertain factors, such as contain dust, density inhomogeneity and irregular shape. In order to carry out the numerical simulation study, some assumptions are necessary to be made as follows:

(1) The thermophysical properties of the structure materials within the de-icing system are independent of temperature;
(2) The ambient temperature and convection heat transfer coefficient are constant and independent of the time;
(3) Perfect thermal contact is assumed to exist between layers;
(4) Ice and water are homogeneous and isotropic. Ignore the volume change in the ice melting process;
(5) The effects of surface curvature are neglected due to the fact that the deicer thickness normal to the airfoil is very small compared to the airfoil chord length;
(6) The phase change phenomena occurs at the temperature of the melting point.

The enthalpy method [31–33] is used in ANSYS Fluent for modeling the melting process. A quantity called liquid fraction (β) is applied to indicate the fraction of the control volume that is in liquid form. The liquid fraction is computed at each iteration based on the enthalpy balance. The mushy zone is defined by the liquid fraction which lies between 0 and 1. When the ice melts completely, the liquid fraction becomes 1. The liquid fraction, β, can be defined as:

Table 1
Physical properties of de-icing system.

Layers	Thickness (mm)	Thermal conductivity (w/m·K)	Specific heat capacity (J/kg·K)
(1)	2.21	115.087	1088.5
(2)	1.27	0.381	962.9
(3)	0.1	13.153	418.7
(4)	0.254	0.381	962.9
(5)	0.305	15.057	489.8
(6)	6.35	–	–

Table 3
Additional computational conditions.

Items	Value
Heater power density	46,500 w/m^2
Ambient temperature	−12.2 °C
Initial temperature	−12.2 °C
Inner convection coefficient	5.68 w/m^2·K
Outer convection coefficient	851.74 w/m^2·K

$$\beta = \begin{cases} 0 & T < T_{melt} \\ 0 \sim 1 & T = T_{melt} \\ 1 & T > T_{melt} \end{cases} \quad (1)$$

The enthalpy of the ice is computed as the sum of the sensible enthalpy, h, and the latent heat, ΔH:

$$H = h + \Delta H \quad (2)$$

Here

$$h = h_{ref} + \int_{T_{ref}}^{T} C_p dT \quad (3)$$

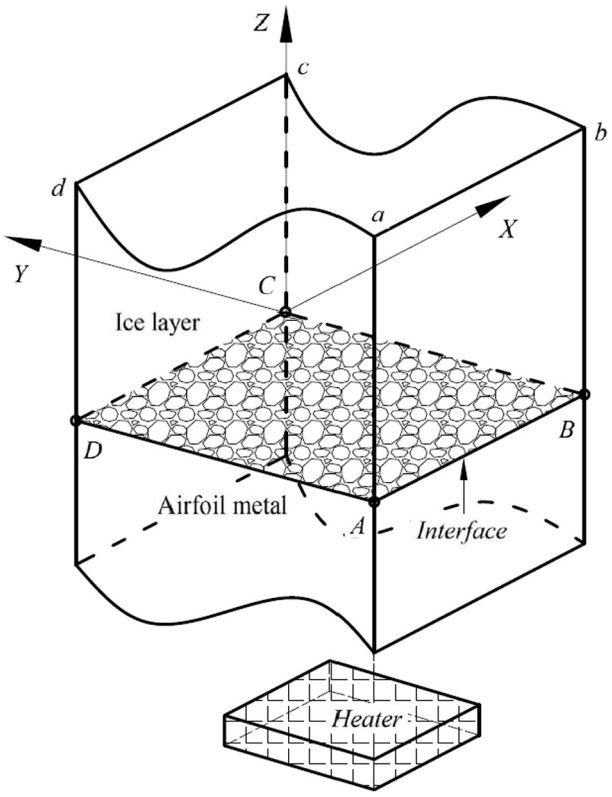


Fig. 3. The sketch of parallel face and data extraction points of three-dimensional de-icing system.

Table 2
Parameters of ice with different density.

Num.	ρ (kg/m^3)	∅ (%)	k_{eff} (w/m-K)	C_{eff} (J/kg-K)	α * 10^6 (m^2/s)
1	400	57	0.9843	1485.421	1.6539
2	500	46	1.2167	1605.537	1.5235
3	600	35	1.4596	1725.653	1.4134
4	700	24	1.7025	1845.768	1.3186
5	800	13	1.9454	1965.884	1.2361
6	920	0	2.2325	2110.023	1.1501

$$\Delta H = \beta L \tag{4}$$

And h_{ref} is reference enthalpy, T_{ref} is reference temperature, C_p is specific heat at constant pressure.

3. Governing equations

The mathematical formulation of unsteady heat transfer in a three-dimensional control volume with heat source is presented as:

$$\rho_j C_{p_j} \frac{\partial T_j}{\partial t} = k_{x_j} \frac{\partial^2 T_j}{\partial x^2} + k_{y_j} \frac{\partial^2 T_j}{\partial y^2} + k_{z_j} \frac{\partial^2 T_j}{\partial z^2} + Q_j \tag{5}$$

where the subscript j represents the layer of interest. This equation represents the energy accumulation rate equaling the heat conduction in three directions plus the local heat generation.

Within the ice layer, the governing equation can be ex-pressed as:

$$\frac{\partial H_{ice}}{\partial t} = \frac{\partial}{\partial x}\left(k_{x_{ice}} \frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k_{y_{ice}} \frac{\partial T}{\partial y}\right) + \frac{\partial}{\partial z}\left(k_{z_{ice}} \frac{\partial T}{\partial z}\right) \tag{6}$$

When the enthalpy of the ice or water is obtained from Eq. (2), the following enthalpy and temperature relationships could be used to determine the temperature:

$$T = \begin{cases} H/C_{ps} & H < H_{sm} \\ T_{melt} & H_{sm} < H < H_{lm} \\ T_{melt} + (H - H_{lm})/C_{pl} & H > H_{lm} \end{cases} \tag{7}$$

where, $H_{sm} = C_{ps} T_{melt}$ and $H_{lm} = C_{ps} T_{melt} + L$.

3.1. Ice parameters

The density of ice is different as the icing condition changs. The density of the accumulated ice is affected by the impinged water droplet diameter, free stream velocity, liquid water content in the free stream and the structure size and surface temperature of the components [34–38]. A random-walk, ballistic-type model was used to simulate the ice accretion process due to impinging supercooled droplets [39]. This model was used to predict ice density under different ambient conditions [40,41]. Macklin developed [34] an empirical ice density formula which used by LEWICE, the most widely used ice accretion simulation program, to calculate the shape of ice accretions. Jones [35,36] indicated that Macklin's density formula did not agree accurately with measurements by the Mt. Washington Observatory in New Hampshire. Through multiple regression analysis, Jones established an ice density formula which can represent the measurement data more accurately. Rios [42] shown that the simulation results using Jones' density formula are in agreement with the experimental ice shapes accreted in the icing research tunnel. Jones' density formula can be expressed as follows [43]:

$$\rho = 1000 \times \exp\left[-0.15 \times \left(1 + \frac{6043}{S^{2.65}}\right)\right]$$
$$S = \frac{MVD^{0.82} \times V^{0.59} \times LWC^{0.21}}{D^{0.48} \times (-T_{wall})^{0.23}} \tag{8}$$

where, MVD is the mean volume diameter with unit μm; V is the free stream velocity with unit m/s; LWC is the liquid water content in the free stream with unit g/m^3; D is the structure equivalent diameter with unit cm; T_{wall} is surface temperature with unit °C.

In this paper, ice is treated as a kind of porous media with different porosity. The pore space is filled with air in the porous media. The porosity is obtained from the follow equation:

$$\phi = \frac{V_{pore}}{V_{total}} \times 100\% \tag{9}$$

where, V_{pore} is pore volume and V_{total} is porous media total volume. It is hard to measure the pore volume but easy to measure the porous media density. So it is better to write the equation of ice porosity as a function of density. For a porous ice, the quality is m kg, the volume is V_{total} m^3 and the density is ρ kg/m^3, respectively. Then:

$$m = \rho \cdot V_{total} = \rho_{pore} \cdot V_{pore} + \rho_{\max} \cdot (V_{total} - V_{pore}) \tag{10}$$

where, $\rho_{\max} = 920$ kg/m^3 is the density for glaze ice with no air in the ice. Then:

$$\frac{V_{pore}}{V_{total}} = \frac{\rho_{\max} - \rho}{\rho_{\max} - \rho_{pore}} \tag{11}$$

So the ice porosity can be expressed as follow:

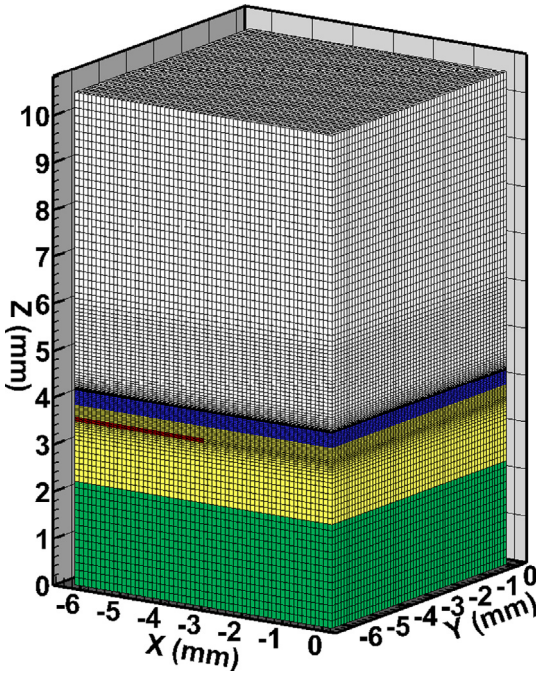


Fig. 4. The numerical simulation mesh.

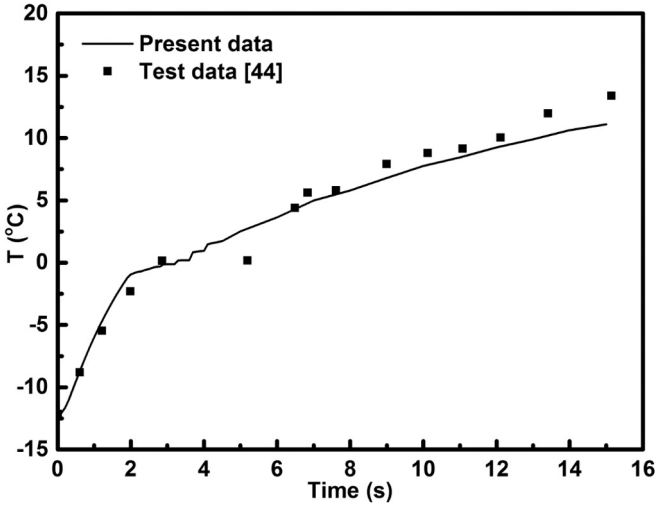


Fig. 5. The comparison of present data and test data.

$$\phi = \frac{V_{pore}}{V_{total}} \times 100\% = \frac{\rho_{\max} - \rho}{\rho_{\max} - \rho_{pore}} \times 100\% \tag{12}$$

Assume that the radiative effect, viscous dissipation, and the work done by pressure changing are negligible and there is a local thermal equilibrium, so $T_s = T_l = T$. T_s and T_l are the temperatures of the solid phase and fluid phase, respectively. Pores are supposed to be distributed evenly in the ice. The effective conductivity k_{eff} is the weighted arithmetic mean of conductivities of k_s and k_l:

$$k_{eff} = (1 - \phi) \cdot k_s + \phi \cdot k_f \tag{13}$$

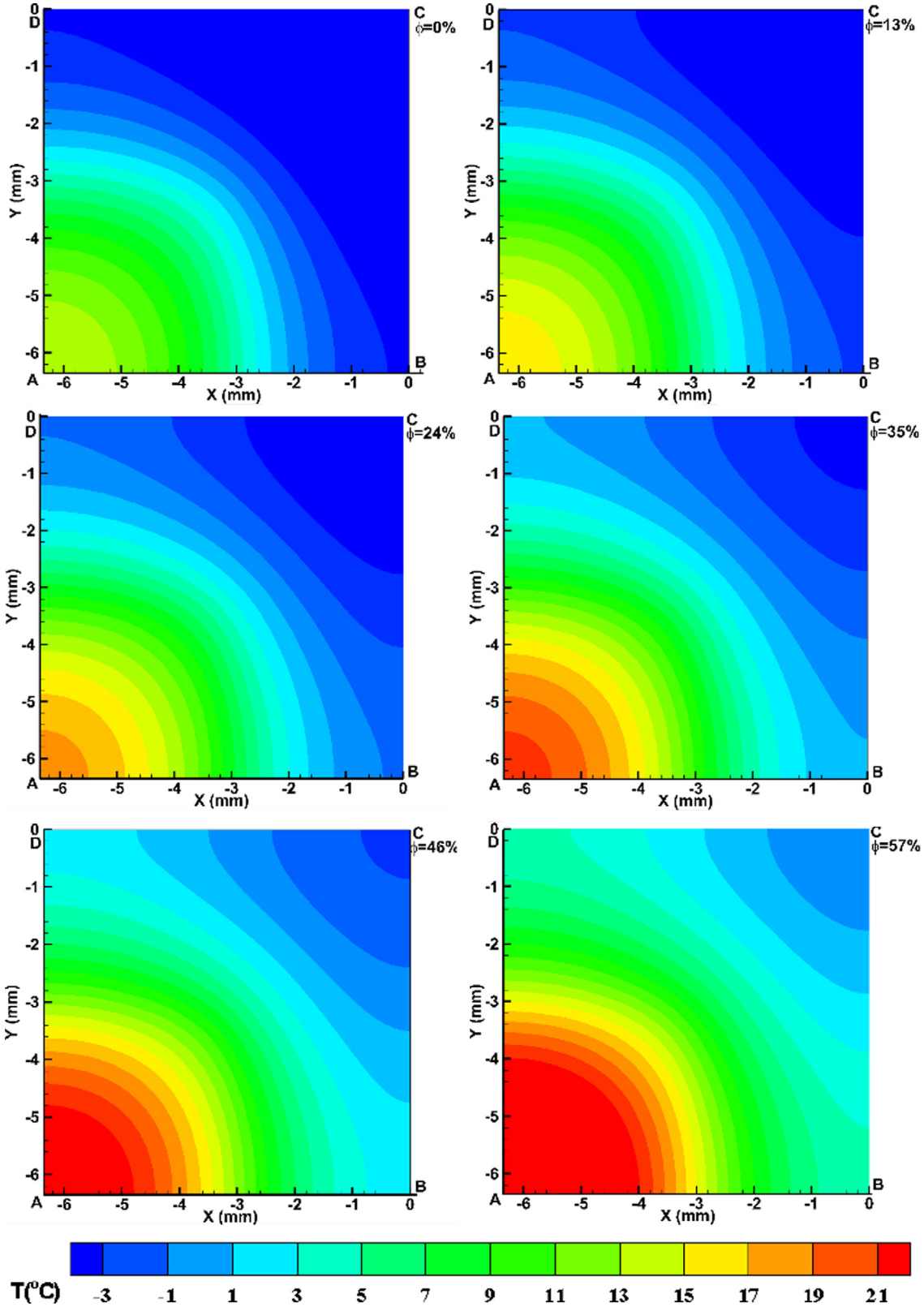


Fig. 6. Temperature distribution of abrasion shield surface at t = 15 s.

The effective heat capacity per unit volume $(\rho C)_{eff}$ is the weighted arithmetic mean of the heat capacities $(\rho C)_s$ and $(\rho c)_f$:

$$(\rho C)_{eff} = (1-\phi)(\rho C)_s + \phi(\rho C_P)_f \quad (14)$$

4. Physical model

The physical model used for simulation is a typical electrothermal de-icing system as shown in Fig. 2. It usually combined with six layers:

(1) Aluminum substrate;
(2) Epoxy/glass insulation;
(3) Heater Element;
(4) Epoxy/glass insulation;
(5) Stainless steel abrasion shield;
(6) Ice.

The layer of aluminum substrate is airfoil structure. The layer of stainless steel abrasion shield is airfoil skin. The layer of epoxy/-glass insulation is used to prevent the airfoil structure and skin from overheated. The material and physical properties of the six layers is shown in Table 1. The ice layer melts when the temperature is above zero. Its physical properties vary with the density.

Six kinds of ice with different density are chosen to analyze the effect of porosity on melting process. The porosity, the effective thermal conductivity and the effective heat capacity are obtained from Eqs. (12), (13) and (14), respectively. The physical properties of ice are given in Table 2.

5. Numerical simulation

5.1. Boundary condition

The geometry of three-dimensional electrothermal de-icing system shown in Fig. 2 is utilized to simulate ice melting process. The heater is 3.175 mm both in length and width. The heater gap length is also 3.175 mm. The boundary condition on the top surface of the ice layer is forced convective with free stream while that on the bottom surface of the aluminum substrate is natural convective with inner air. Symmetry boundary is applied to the vertical faces. The numerical simulation conditions are given in Table 3. The heater power density is 46500 W/m^2 and turns on for 15 s.

In order to trace the ice melting process and the transient temperature variation in the melting zone, a face and four points are identified in Fig. 3. The abrasion shield surface, which contact with ice, is defined as the interface as shown in Fig. 3. The line A-a passes through the center of the heater. The lines B-b and D-d pass through the center of the gap. The line C-c is farthest away from the heater. These lines are perpendicular to the heater and the interface, and produced four intersection points (A, B, C, D). The point A is above the heater directly and has the shortest distance with the heater. The point B and the point D have the same distance relative to the heater. The point C is the farthest point away from the heater. The temperature of point A will be the highest and the temperature of point C will be the lowest when the heater works. The ice melts on the place of the point A first and melts on the place of the point C finally. The point C is the key position to judge whether the ice melts completely.

5.2. Computational mesh

The structural hexagon mesh is applied in simulations. In order to avoid the effect of the grid on the numerical simulation results, the grid independence has been verified. Refined grid is used in boundary layers. The computational mesh is shown in Fig. 4. The boundary layer grid is created on the interface of the ice and abrasion shield surface.

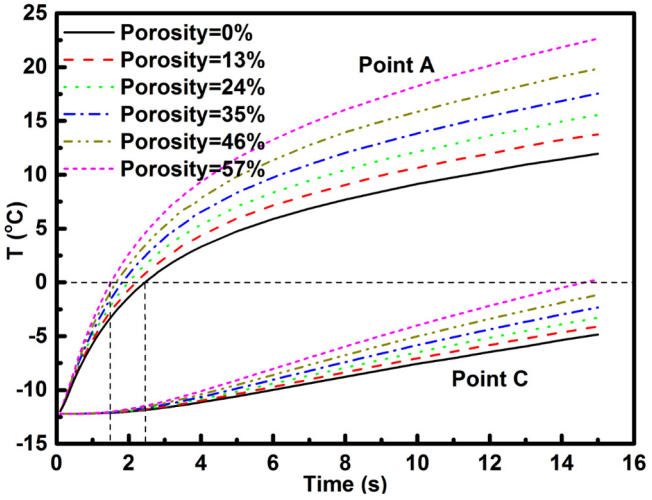


Fig. 7. Transient temperature curves of point A and C.

5.3. Model validation

The transient temperature of glaze ice melting process was obtained from experimental in reference [44]. The glaze ice porosity equals 0%. A numerical simulation is carried out to simulate the glaze ice melting process. The transient temperature at point A compared with reference [44] are shown in Fig. 5. The black line indicates the numerical simulation transient temperature of point A. The black squares indicate the experimental transient tempera-

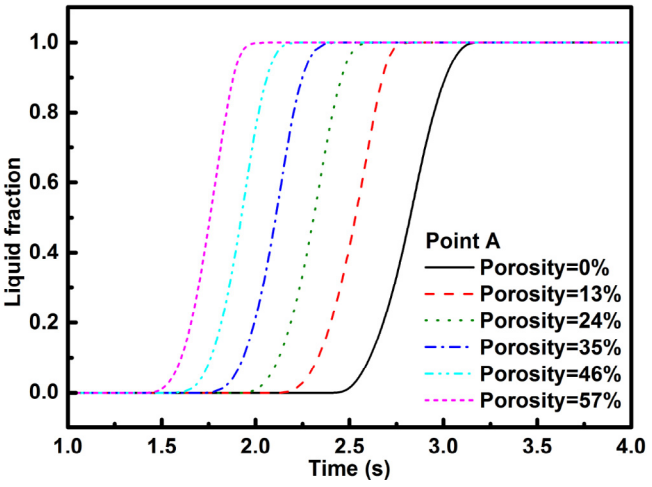


Fig. 8. Transient liquid fraction curves of point A.

Table 4
The melting time of point A with different porosity.

Porosity (%)	Stat time (s)	Complete time (s)	Duration time (s)
0	2.5	3.1	0.6
13	2.3	2.8	0.5
24	2.1	2.5	0.4
35	1.9	2.3	0.4
46	1.7	2.1	0.4
57	1.6	2.0	0.4

ture around the point A in reference [44]. The main difference between the numerical result with the experimental result appears when the ice starts melting but not melted completely. At that moment, the melting zone is an ice-water mushy zone. The temperature of the ice-water mushy zone is zero as shown in the experimental result. In the numerical simulation, the point A is on the abrasion shield surface. The comparison results show that the model used in this paper is reliable.

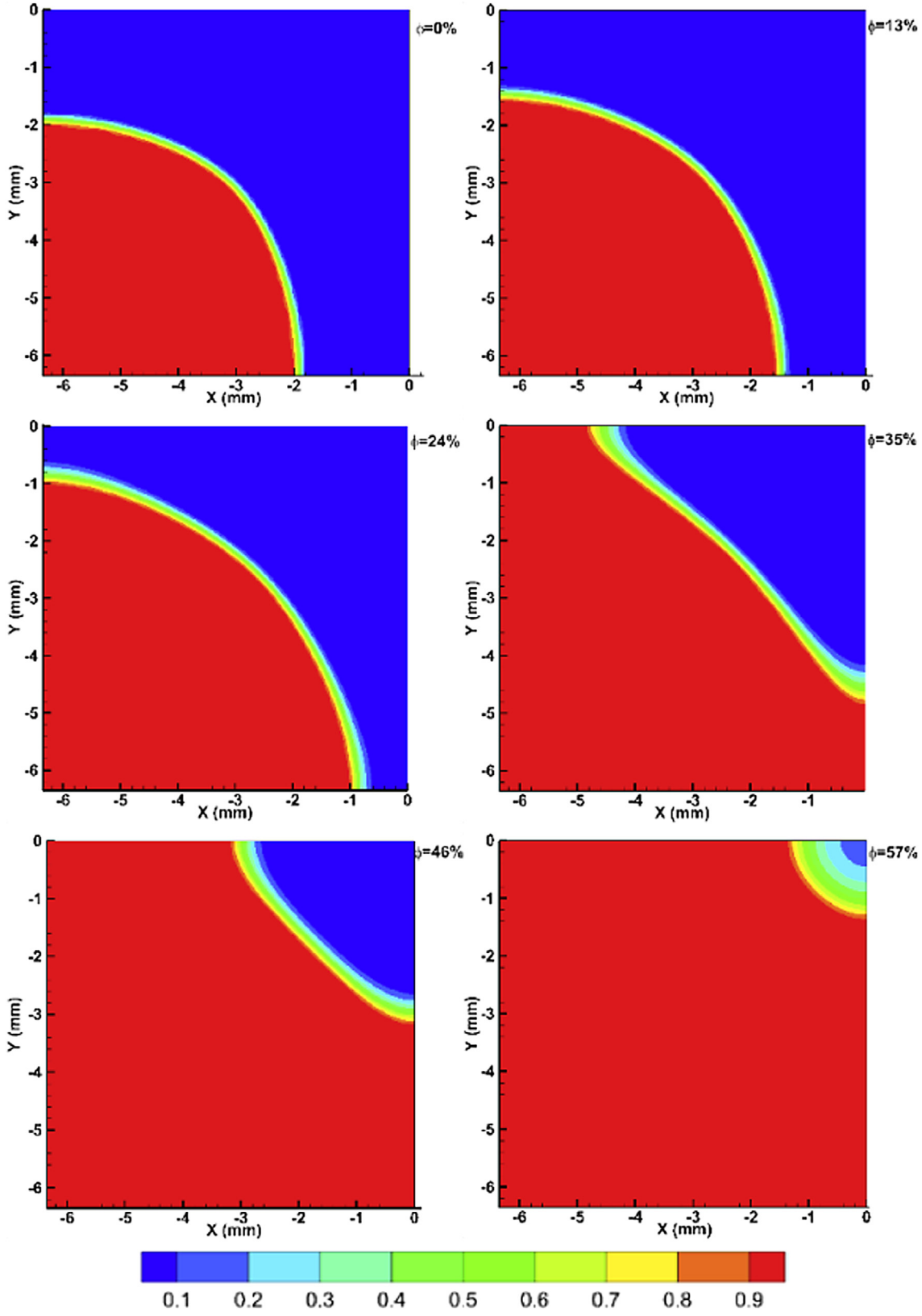


Fig. 9. Liquid fraction on the abrasion shield surface at t = 15 s.

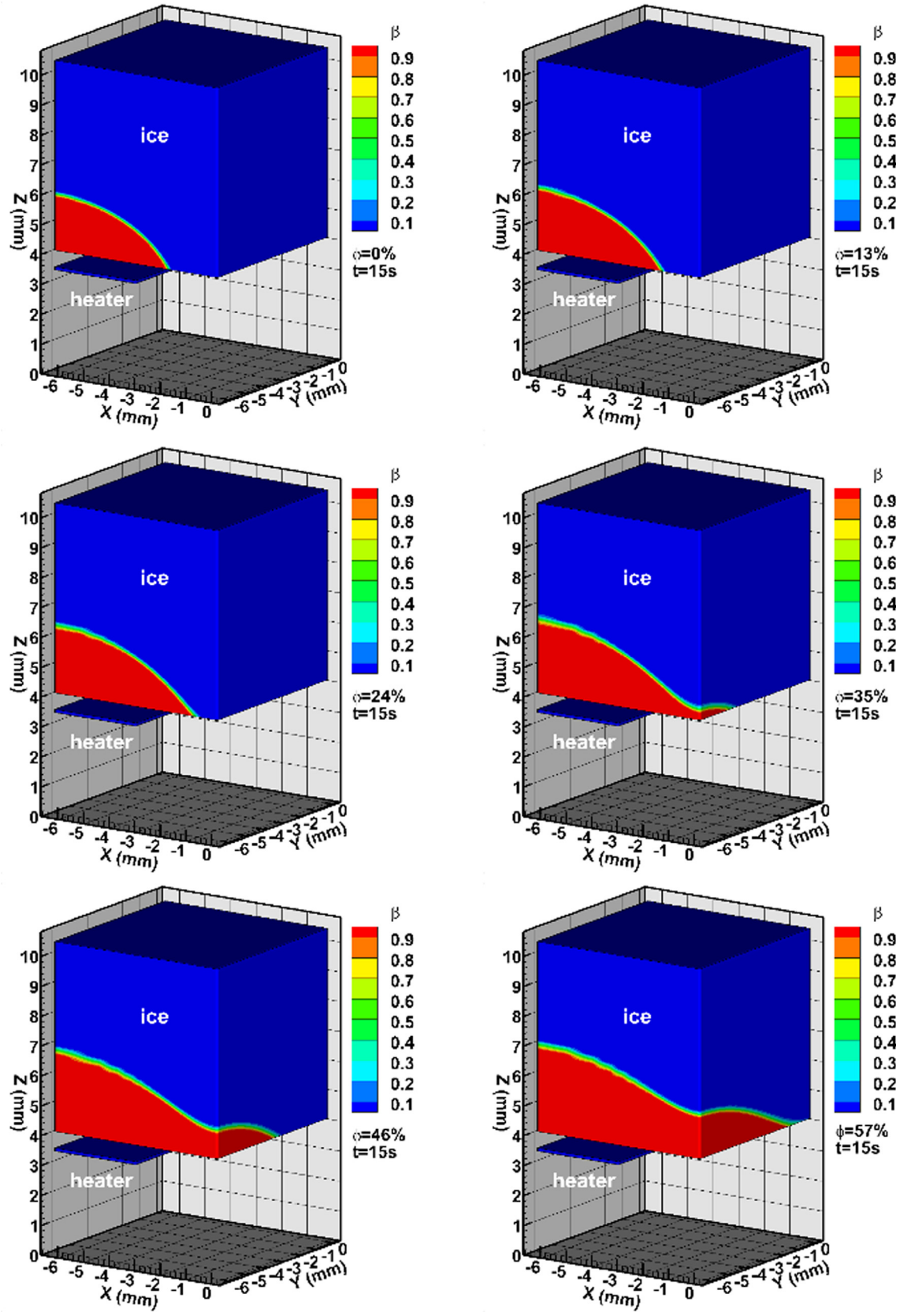


Fig. 10. The melted thickness and volume at t = 15 s for the ice with different porosities.

6. Results and discussion

In order to investigate the effect of porosity on the melting temperature, the melting rate, the melted region, the melted volume and the natural convection in the melted zone, numerical simulations were carried out based on the three-dimensional de-icing system model shown in Fig. 3. The ice initial temperature is -12.2 °C and the heater works 15 s continuously.

6.1. The effect of porosity on temperature distribution

The Fig. 6 shows that the temperature distribution on the interface at 15 s after heating. The highest temperature occurs at point A and the minimum temperature occurs at point C. The temperature contour shows that the interface temperature rises with the increase of the ice porosity. The maximum temperature on the abrasion shield surface is about 13 °C for ice with porosity of 0% while it is about 23 °C for ice with porosity of 57%. The interface temperature distribution changes with the changing of the ice porosity. The interface temperature ranges from −3 °C to 13 °C for ice with porosity of 0% while it ranges from 0 °C to 23 °C for ice with porosity of 57%. The results show that porosity affects the ice temperature distribution significantly. The porosity affects the effective thermal conductivity and the effective heat capacity of the ice. Both of the effective conductivity and the effective heat capacity of the ice decrease with the increase of the porosity. The ice thermal diffusivity increases with the increasing of the porosity.

The transient temperatures of points A and C for ice with different porosities are shown in Fig. 7. The temperature rises with the porosity increase at the same time after the heater working. The lowest temperature occurs at point C, the ice on the abrasion shield surface melts totally when the temperature of point C is above 0 °C. The Fig. 7 shows that the temperature of point C reached 0 °C for ice with porosity 57% when heater works for 15 s. So the ice can be removed for the case of ice with porosity 57%. The de-icing time reduces with the increase of porosity. The heating time of raising the temperature of point A to above 0 °C needs 2.5 s for ice with porosity of 0% while it needs 1.5 s for ice with porosity of 57%. The de-icing system need to work for long time to remove the ice completely for lower porosity ice. The glaze ice with porosity of 0% start to melts at pint A when the heater works 2.5 s, but the temperature of point C is still under 0 °C.

6.2. The effect of porosity on melting rate

The liquid fraction at points A for ice with different porosities is shown in Fig. 8. The liquid fraction changed from 0 to 1 when the ice melted to water. The porosity affects the ice melting process, especially the time of ice starts to melt and the time of ice melts completely. Fig. 8 shows that the point A melting into water starts from 2.5 s and melts completely at 3.1 s when the ice porosity equals 0% while it starts from 1.6 s and completely at 2.0 s when the ice porosity equals 57%.

The melting time of point A of ice with different porosity is shown in Table 4. For the ice with different porosities, the ice melting starts at an earlier time with the porosity increase. The duration time of the ice melting process reduces when the porosity increases. The whole melting process spent about 0.6 s for the ice with porosity of 0% while it spent about 0.4 s for the ice with porosity of 57%. The melting time reduces by 33.3% at point A when the ice porosity increases from 0% to 57%. The ice melting rate speeds up significantly as the ice porosity increases.

6.3. The effect of porosity on melted region

Fig. 9 shows the contours of the liquid fraction on the interface when the heater works 15 s. The red zone means that the ice has melted into water and the blue zone means that the ice has not started to melt. These figures demonstrate that the ice melted region on the interface enlarges with the porosity increase. The ice melted region enlarges from about 14.9 mm^2 to about 38.9 mm^2 when the ice porosity increases from 0% to 57%. The area of melted region increased by 161%. The porosity has a great influence on the ice melting region on the abrasion shield surface as shown in Fig. 9.

6.4. The effect of porosity on melted volume

Fig. 10 shows the melted zones of ice with different porosities after the heater working for 15 s. These figures show that the thickness and volume of the melted zone increases with the increase of porosity. The reason is that the ice effective thermal diffusivity increases with the increase of the porosity.

The maximum melted thickness along the line A-a of the ice with different porosities at different time is shown in Fig. 11. The melted thickness increases along with the melting process goes on. And the melted thickness increases with the increase of porosity at the same heating time. The ice melted thickness increases from 1.8 mm to 2.7 mm when the ice porosity increases from 0% to 57%.

6.5. Buoyancy-driven flow

Fig. 12 shows the effect of porosity on buoyancy-driven flow in ice melting zone after 15 s heating. The figure shows the streamlines in melted zone of ice with different porosities. The natural convection phenomenon occurs in the melted region when the ice melts. The intensity of the natural convection strengthens as the increasing of melted region. The natural convection enhances the heat transfer and increases the ice melting rate. In the melted zone, the water heated by abrasion shield surface rises to the top and replaces the cold fluid. The cold water flows along the ice-water interface and arrives at the abrasion shield surface. The natural convection transfers the heat from high temperature zone to low temperature zone. The flow velocity and the natural convection heat transfer will increase with the increasing of the temperature and the volume of the melted zone.

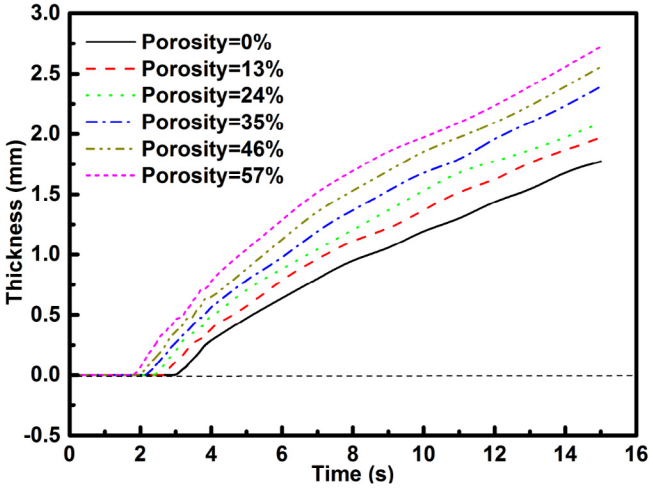


Fig. 11. The thickness of melted ice zone with different porosities.

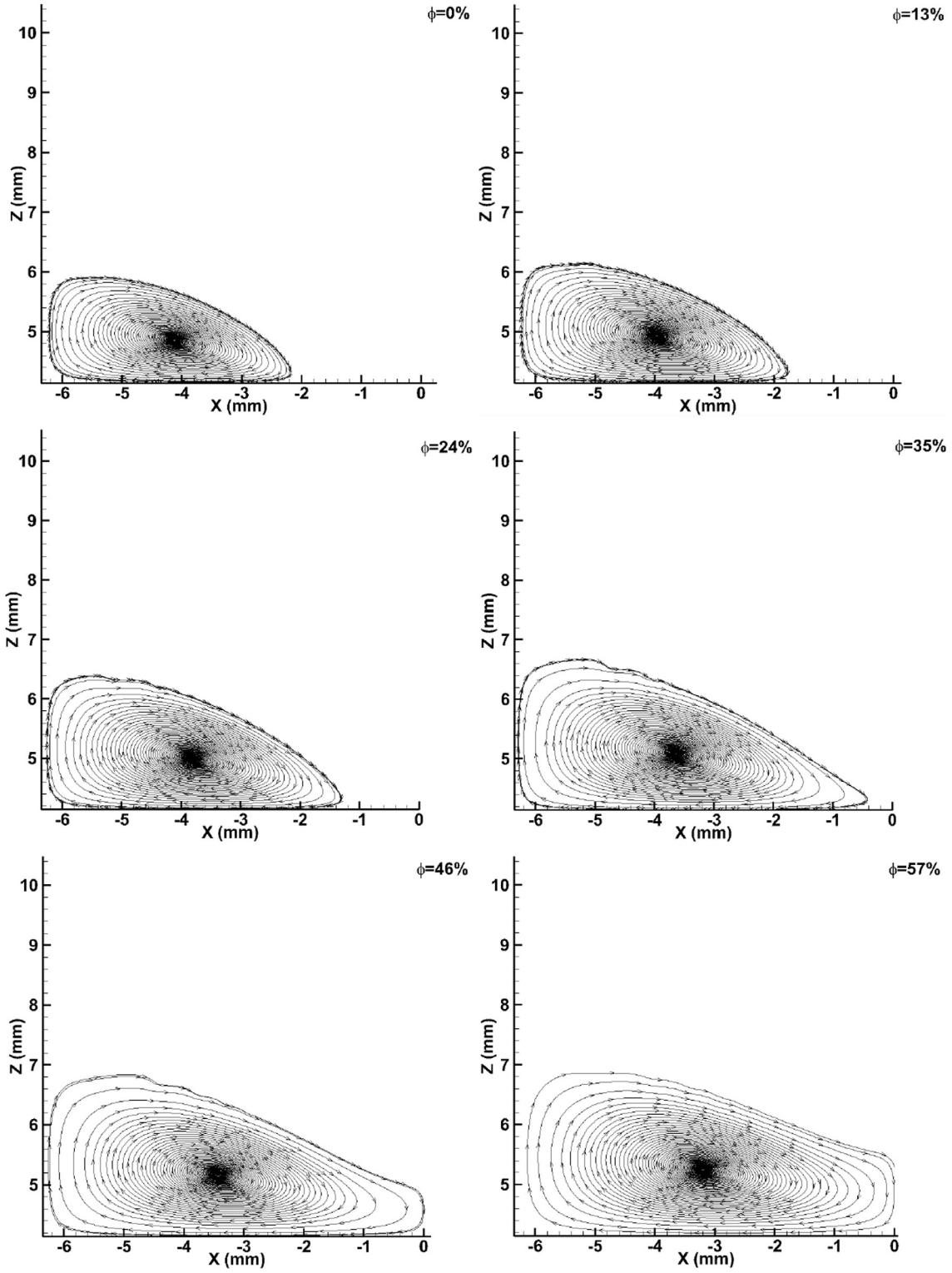


Fig. 12. The streamlines of melted zone with different porosities at t = 15 s.

7. Conclusions

Ice density is different under different icing conditions. Ice can be treated as a porous medium with different porosities. A melting model based on the enthalpy-porosity method is applied to study the effects of porosity on the ice melting process and heat transfer. An electrothermal de-icing system is used to investigate the melting process of ice with different porosities. The transient process of ice melting is investigated, including the change of ice-water interface, the melted ice volume and the melted ice area on the surface. The effects of porosity on the temperature distribution on the abrasion shield surface are analyzed. The effect of natural convection on ice melting process is also considered in this study. The results show that the porosity affects the temperature distribution of the

ice melting zone, melting rate, melting ice area on the surface and melting ice volume significantly. The melting rate, melting ice area on the surface and melting ice volume in the ice melting process are all rise with the increase of porosity under the same heating condition. The investigation shows that the enthalpy-porosity method can describe the ice melting process more accurately.

The increase of porosity strengthens the natural convection in the melting zone, and the natural convection can increase the ice melting rate. The porosity affects the effective thermal conductivity coefficient and the effective heat capacity of the ice as well. Both of the effective thermal conductivity coefficient and the effective heat capacity decrease with the increase of the porosity.

The investigation in this paper is beneficial to improve the understanding of the mechanism of the ice melting process and is helpful to the design of the electrothermal de-icing system.

Acknowledgements

The author would like to thank the National Basic Research Program of China (973 program) under Grant No. 2015CB755800 and the National Natural Science Foundation of China under Grant No. 11272212 and No. 11572195.

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